

VRUTANT SEMRAJ KAR

AI & Machine Learning Engineer | Agentic AI | Software Developer | Computer Vision

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GitHub: [github.com](https://github.com/vrutant) | Portfolio: [portfolio-vrutant](https://portfolio-vrutant.vercel.app/) | Location: Nashik, Maharashtra

PROFESSIONAL SUMMARY

Aspiring AI & Software Developer with a strong foundation in **Machine Learning, Agentic AI, and Software Development**. Passionate about building intelligent, autonomous systems that solve real-world problems. Hands-on experience with **n8n workflow automation**, designing multi-step agentic pipelines, and integrating AI models into scalable applications. Eager to leverage cutting-edge tools and frameworks like **LangChain, Python, and LLMs** to contribute to innovative AI-driven solutions.

TECHNICAL SKILLS

Languages : Python, Java, C++
Web : HTML5, CSS3
Databases : SQL, MySQL
Tools & IDEs : Git, GitHub, VS Code, n8n
OS : Windows
Concepts : OOP, Data Structures, REST APIs, SDLC

EDUCATION

Bachelor of Engineering — Computer Science (AI & ML)

Sandip University, Nashik | 2023–2027 (*Expected*) | CGPA: **8.2 / 10**

Higher Secondary Certificate (12th) — Computer Science (PCM)

N. H. High School, Bramhapuri | 2022 | Percentage: **71.83%**

Secondary School Certificate (10th) — CBSE

L.M.B. Public School, Bramhapuri | 2020 | Percentage: **82%**

PROJECTS

Smart Traffic Signal Control System | Python, JavaScript, Machine Learning [GitHub Link](#)

- Developed an AI-powered smart traffic signal control system using **Machine Learning** to dynamically optimize signal timing based on real-time traffic patterns.
- Built a full-stack architecture with a **Python backend** (ML model + logic) and **JavaScript frontend** for live traffic monitoring and control dashboard.
- Implemented a **traffic simulation module** to test and validate ML model performance under various traffic load scenarios.
- Automated CI/CD workflows using **GitHub Actions** for continuous integration and deployment.

Loan Approval Prediction System | Python, Machine Learning, Scikit-learn

(*Jupyter Notebook — ML Classification Project*) [GitHub Link](#)

- Built a **binary classification ML model** to predict loan approval outcomes using applicant financial and demographic data.
- Performed end-to-end **data preprocessing** including missing value imputation (mode/median), label encoding, and one-hot encoding for categorical features.
- Trained a **Random Forest Classifier** with hyperparameter tuning (max_depth, max_features, max_samples) to optimize prediction accuracy.
- Evaluated model performance using **Confusion Matrix, Accuracy Score, Classification Report, and ROC Curve**.
- Visualized **feature importances** using Matplotlib to identify key factors influencing loan approval decisions.

INTERNSHIPS / TRAINING

AI & DS Training and Internship — Intellipaat, Online (*March 2026 – Ongoing*)

- Attended daily 1.5-hour live instructor-led sessions covering core Data Science concepts and applications.
- Completed multiple hands-on assignments to reinforce practical understanding of Data Science tools and techniques.
- Gained applied exposure to real-world Data Science use cases through structured coursework.

Data Science Training & Internship — Dezinolics Web & Software Company (*1 Month*)

- Completed a 7-day training program on Data Science fundamentals, tools, and applications.
- Underwent a 21-day internship applying Data Science concepts to practical/real-world tasks.

- Gained hands-on exposure to the end-to-end Data Science workflow under industry mentorship.

CERTIFICATIONS

- **Generative AI for Beginners** — Great Learning Academy (Nov 2024)
- **Introduction to Machine Learning** — Great Learning Academy (Nov 2024)
- **Artificial Intelligence Fundamentals** — Great Learning Academy (Nov 2024)
- **Python Mastery for Beginners** — Udemy, The Kiran Academy (Oct 2025)
- **AI Tools Workshop** (AI Tools & ChatGPT) — be10x (Jan 2026)
- **HackX Hackathon | Certificate of Participation** — Certify X (Jan 2026)
- **Data Science** — Certificate of Internship (June – July 2026)

LANGUAGES

English (Professional) | Hindi (Fluent) | Marathi (Native)